

Richard Tang

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science Major, concentration in AI

May 2028

- Relevant Courses: Algorithms and Data, Fundamentals of Computer Science 1&2, Object Oriented Design, Discrete Structures, Matrix Methods in Data Analysis and Machine Learning.
- Honors: GPA: 3.96/4.0, Dean's List, Dean's Scholarship.
- Activities: AeroNU, Robotics NU.

EXPERIENCE

Machine Learning Engineer

January 2026 – Present

Infinite Cooling (MIT-affiliated startup)

Malden, MA

- Architected agentic RAG applications with multi-query hybrid retrieval and custom tool orchestration, reaching 53.8% judged acceptance on internal benchmark (+24.7 pts vs GPT-5.2, +85% relative).
- Orchestrated backend operational data integration into the agent, enabling natural-language querying of metrics and performance summaries.
- Developed automated anomaly-detection software and sensor alerts, improving prior runtime efficiency by 12x.
- Leveraged DOE chiller-model time-series data for anomaly detection and operation optimization.

Software Engineer

September 2025 – Present

AeroNU (Northeastern University)

Boston, MA

- Engineered a rocket-detection framework using YOLOv26 model and classical computer vision algorithm, achieving 84% mAP on internal benchmarks while operating at 30+ frames per second.
- Deployed detection framework on Jetson-Nano with active gimbal to support rocket tracking in real-time.
- Constructed a Webots simulation environment to generate large scale training/evaluation data for object detection algorithms used on drone deployment.

Undergraduate Researcher

September 2024 – Present

Northeastern University

Boston, MA

- Enhanced Qwen2.5-VL 3B with advanced spatial reasoning using VGGT as a 3D geometry encoder; applied LoRA and supervised fine-tuning on specialized datasets for robust 3D scene understanding.
- Scaled and optimized model training and fine-tuning on a high-performance computing cluster.
- Investigated geometric and visual embedding fusion across different modes of data and geometric encoders.

PROJECTS

Spatial Reasoning for VLLMs

October 2025 – Present

- Developed multimodal memory designs for spatial awareness using Qwen2.5, achieving 48.3% on VSI-bench
- Built PyTorch distributed evaluation framework with multi-GPU inference to evaluate experimental designs
- Designed pose-based keyframe sampling, and extended custom Qwen2.5-VL memory model to reason over geometric information

Autonomous Spatial Tagging and Reconnaissance framework

August 2025

- Designed a real-time object tracking system using a client-server architecture to reduce onboard computation.
- Integrated a YoloV8 and SORT-based object tracking module to identify and maintain consistent tracking.
- Built a 3D triangulation module for detected objects using feature matching and bundle adjustment techniques.

Ming DeepResearch

February 2025

- Engineered a multi-agent deep-research system using knowledge-graphs for cost-effective context management.
- Achieved competitive performance on DeepResearch Bench, ranking 4th amongst proprietary and open-source services.
- Scaled the agentic system to deliver at \$ 0.72 per-run, 90% cheaper compared to proprietary services

TECHNICAL SKILLS

Languages: Python, C++, Java, Go, SQL

ML/AI Frameworks: Pytorch, TensorFlow, Scikit-learn, Huggingface, LangChain, LangGraph

Tools & Others: Pandas, Numpy, Docker, Flask, Fastapi, Supabase, Matplotlib